

SHERWIN TANG

AI / Applied Machine Learning Engineer | Full-Stack Software Engineer
Singapore • sherwintang93@hotmail.sg • +65 9477 7959 • [LinkedIn](#) • [GitHub](#)
Singapore Citizen

PROFESSIONAL SUMMARY

Software engineer with 7+ years of professional experience, including 4+ years building and shipping production-grade systems at scale, now specializing in applied AI, machine learning, and Generative AI. Combine strong full-stack and platform engineering (microservices, Kubernetes, CI/CD) with hands-on experience designing and integrating LLM/RAG and reasoning-based AI systems through an NUS MTech in Artificial Intelligence Systems. Proven at taking ideas from prototype to production through rapid POCs, end-to-end integration, and cross-functional, cross-border delivery. Keen interest in explainable, reliable AI for real-world, safety-critical applications.

TECHNICAL SKILLS

AI / Machine Learning: LLMs & Generative AI (OpenAI API), Retrieval-Augmented Generation (RAG), LangChain, Hugging Face Transformers, sentence-transformer embeddings, Pinecone / vector databases, neuro-symbolic reasoning, knowledge graphs, rule-based systems, informed search & planning, NLP, computer vision, PyTorch, TensorFlow, scikit-learn, Pandas

Languages: Python, Java, JavaScript, HTML, CSS

Web & Rapid Prototyping: React.js, Streamlit, Spring / Spring Boot, REST APIs

Cloud & DevOps: AWS (S3), Kubernetes, OpenShift, Helm, CI/CD (SonarQube, BlackDuck), Git, Apache Kafka, Control-M, Unix

Databases: Oracle, PostgreSQL

Practices: Microservices architecture, Agile, Scrum, test-driven & integration testing

AI / MACHINE LEARNING PROJECTS

WellnessBot — Rehabilitation Decision Support System

2026

NUS MTech in AI Systems • Intelligent Reasoning Systems

A home-based, explainable clinical decision-support assistant for post-arthroscopic knee rehabilitation. Built on a neuro-symbolic architecture (knowledge graph + rule-based reasoning) with a constrained RAG/LLM layer, so that all clinical decisions remain deterministic, transparent, and auditable rather than left to an opaque LLM.

- Designed and implemented the informed search planner and evaluated alternative search strategies to generate safe, rehabilitation-phase-appropriate recommendations.
- Orchestrated end-to-end pipeline integration across the knowledge graph, rule engine, RAG module, and LLM; owned system integration and end-to-end debugging.
- Built the user interface (Streamlit) and translated structured system decisions into clear natural-language responses for patients.
- Implemented error handling, edge-case management, and fallback logic, and led integration testing to ensure reliable, predictable behavior.
- **Stack:** Python, OpenAI API, RAG, knowledge graph, rule-based reasoning, Streamlit. [Live demo](#) • [GitHub repo](#)

Retrieval-Augmented Generation (RAG) Q&A System

2025

NUS MTech in AI Systems • RAG Workshop

An end-to-end RAG pipeline built with LangChain that answers document- and FAQ-grounded user queries (course enrolment, schedules, funding eligibility) from a custom knowledge base, grounding every response in retrieved source content to reduce hallucination.

- Ingested and chunked source documents (PDF and CSV) using LangChain document loaders and a recursive character text splitter.
- Generated embeddings with a Hugging Face sentence-transformers model (all-mpnet-base-v2) and indexed them in a Pinecone vector store using cosine-similarity retrieval.
- Assembled a LangChain RetrievalQA chain to generate grounded answers over the top-k retrieved passages.
- Integrated and compared two interchangeable LLM backends — a local Hugging Face Transformers model (TinyLlama-1.1B-Chat) and the OpenAI API (GPT-3.5) — weighing cost, latency, and answer quality.
- **Stack:** Python, LangChain, Hugging Face Transformers & sentence-transformers, Pinecone, OpenAI API.

PROFESSIONAL EXPERIENCE

Software Engineer (Trade Finance Squad) — Team Lead

Oct 2021 – Present

OCBC Bank, Singapore

- Led the migration from a monolithic architecture to microservices, improving system efficiency by ~50% and reducing operational costs by ~20%.
- Managed a cross-border development team (Singapore & China) across 20+ microservices using Agile, increasing project delivery throughput.
- Partnered with Product Owners, UX Designers, and Business Analysts to translate new business and regulatory requirements into delivered solutions with a 100% compliance record.
- Spearheaded the launch of Trade Finance services in Vietnam, enabling online applications and increasing transaction volume by ~20%.
- Led development and deployment of the Export Documentary Collection product, introducing a fully online application channel and substantially increasing adoption.
- Maintained and optimized Trade Finance microservices through unit testing, legacy-issue resolution, and production deployments, improving system reliability.
- Executed technical upgrades for system resiliency, including Kafka broker upgrades, data-migration exercises, and disaster-recovery (DR) simulations.
- Delivered key enhancements: redesigned web interfaces and APIs for simpler document resubmission; improved transaction-status workflows and notifications; regionalized products across APAC; integrated third-party SSO for a smoother authentication flow.
- Worked across databases, data processing, and data analysis, with CI/CD tooling (SonarQube, BlackDuck), Kafka, Kubernetes, and Control-M.

Senior Executive — Strategic Planning & Digital Transformation

Jul 2018 – Jul 2021

SMRT Trains Ltd, Singapore

- Led development and User Acceptance Testing (UAT) of a Kaizen engagement portal, significantly increasing employee-driven process-improvement initiatives.
- Built a Power BI dashboard giving senior management real-time visibility into employee feedback and initiatives.
- Directed digitalization of maintenance approval workflows, reducing approval time from 6 days to 1 day and cutting paper usage by ~75%.
- Ensured KPI accuracy in regulatory reports and created data-driven dashboards for management insight.

EDUCATION

Master of Technology in Artificial Intelligence Systems (Part-time)

Jul 2025 – Present

National University of Singapore

- Core modules: Intelligent Reasoning Systems, Pattern Recognition Systems.
- Hands-on experience with ML algorithms, neural networks, computer vision, and NLP through applied projects (incl. WellnessBot above).

B.Eng. (Hons) in Materials Science & Engineering

Aug 2014 – May 2018

National University of Singapore

CERTIFICATIONS (NATIONAL UNIVERSITY OF SINGAPORE)

- [Cognitive Systems](#) — Apr 2026
- [Reasoning Systems](#) — Apr 2026
- [Machine Reasoning](#) — Mar 2026
- [Intelligent Sensing and Sense Making](#) — Nov 2025
- [Pattern Recognition and Machine Learning Systems](#) — Nov 2025
- [Problem Solving using Pattern Recognition](#) — Sep 2025

LANGUAGES

English (Native) • Chinese (Native)